Lesson 1 RGB Color Control

This section demonstrates controlling the RGB light color on the expansion board. For example, sets the color of it to be red and green.

1. Working Principle

RGB represents red, green, blue three color channels. Their parameter ranges are all from 0 to 255, with higher value indicating darker color. In the program, the flashing frequency of the RGB lights is controlled by modifying the delay time for turning them on and off. The source code of the program is located in : /home/pi/board_demo/rgb_control_demo.py

```
while True:
    #设置2个灯为红色(set two RGB lights to red)
    board.set_rgb([[1, 255, 0, 0], [2, 255, 0, 0]])
    time.sleep(1)

    #设置2个灯为绿色(set two RGB lights to green)
    board.set_rgb([[1, 0, 255, 0], [2, 0, 255, 0]])
    time.sleep(1)

    #设置2个灯为蓝色(set two RGB lights to blue)
    board.set_rgb([[1, 0, 0, 255], [2, 0, 0, 255]])
    time.sleep(1)

    #设置2个灯为黄色(set two RGB lights to yellow)
    board.set_rgb([[1, 255, 255, 0], [2, 255, 255, 0]])
    time.sleep(1)

    if not start:
        #所有灯关闭(close all RGB lights)
        board.set_rgb([[1, 0, 0, 0], [2, 0, 0, 0]])
        print('已关闭(closed)')
        break
:set number                                          44,38          Bot
```

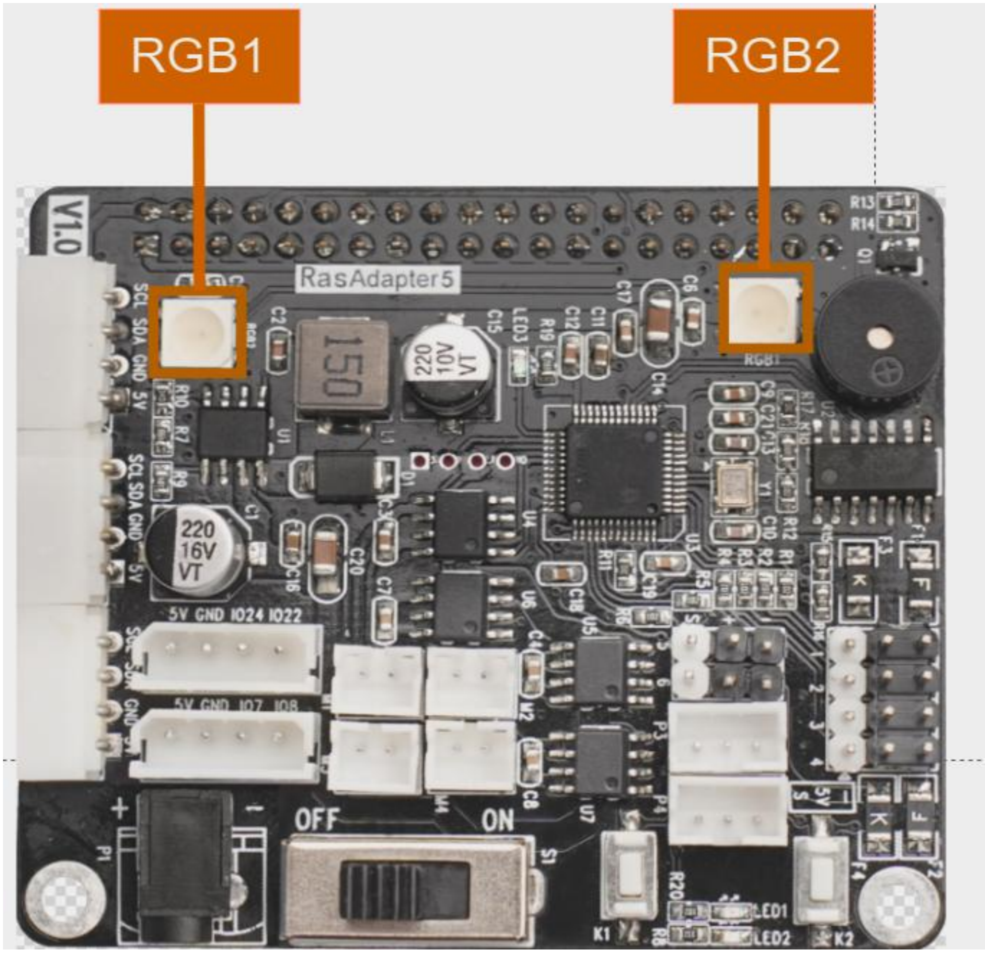
① This section demonstrates controlling RGB through calling the "set_rgb()" function in the Board library. Take "board.set_rgb([[1, 255, 0, 0], [2, 255, 0, 0]])" as an example:

② Among that, the first parameter "1" means RGB2. Parameter "2" represents RGB2.

③ The second parameter “255,0,0” represents color channel parameter. Among that, the first parameter “255” represents the value of “R” channel (red); the second and the third parameters represent the value of “G”, “B” channel respectively.

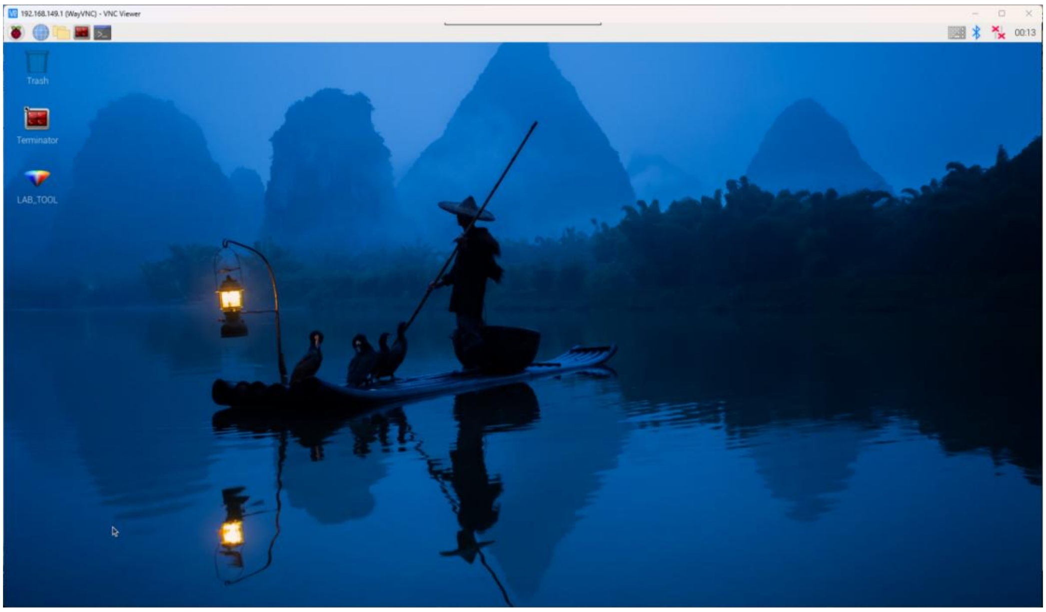
2. Getting Ready

The Raspberry Pi expansion board has two RGB lights as pictured:

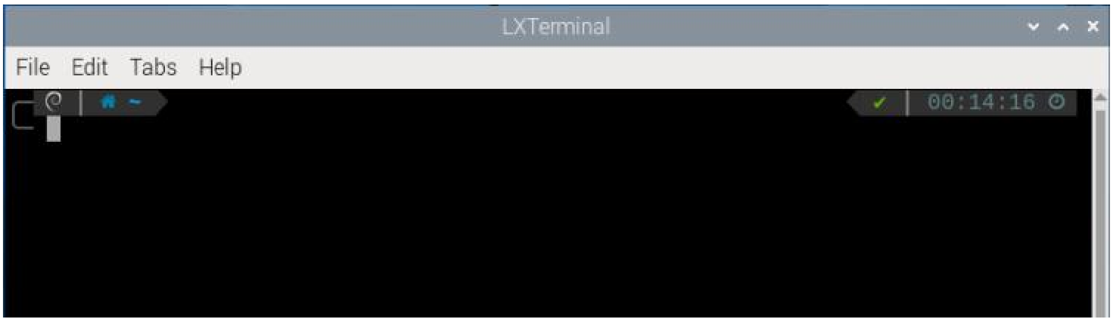


3. Operation Steps

1)Turn the device on and connect the robot through the VNC remote connection tool.



2)Click the icon  in the left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



3)Enter the command “cd board_demo/” in the directory where the program is located.

4)Next, enter the command “python3 rgb_control_demo.py” to run the program.

```
pi@raspberrypi:~/board_demo $ python3 rgb_control_demo.py

**********************************************************
********功能:幻尔科技树莓派扩展板，RGB灯控制例程(function:Hiwonder Raspberry Pi
expansion board, RGB LED control)**********
**********************************************************
----------------------------------------------------------
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
----------------------------------------------------------
Tips:
 * 按下Ctrl+C可关闭此次程序运行，若失败请多次尝试！(Press "Ctrl+C" to exit the p
rogram. If it fails, please try again multiple times!)
----------------------------------------------------------
```

5) You can press “Ctrl+c” to close this program. If you fail to close the program, repeat the process until you exit.

4. Program Outcome

Once the program is running, the two RGB lights on the Raspberry expansion board emit red, green, blue and yellow in a loop.